

- 2 Interferon-alpha (IFN- α) can be produced as a recombinant human protein to treat some types of cancer. The gene *IFNA2* codes for IFN- α .

One method of producing recombinant IFN- α uses genetically engineered *Escherichia coli* bacteria that contain recombinant plasmids. Each recombinant plasmid contains:

- the gene *IFNA2*
- three regulatory sequences of the *lac* operon (promoter, operator and *lacI*)
- a gene for antibiotic resistance, *AMP^R*.

Each of the sequences for the *lacI* gene and *AMP^R* gene contains its own promoter. As a result, these genes are always expressed in *E. coli* bacteria that contain this recombinant plasmid.

Fig. 2.1 is a diagram of the recombinant plasmid. The promoter regions of the *lacI* gene and *AMP^R* gene are **not** shown.

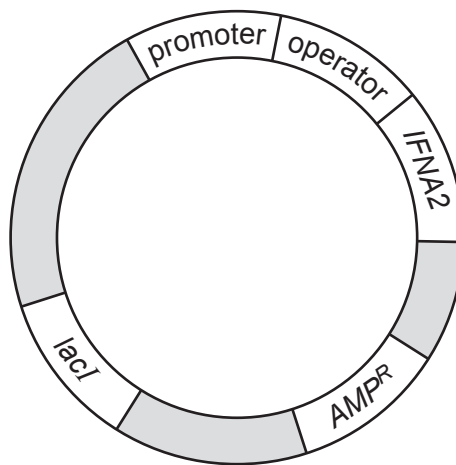


Fig. 2.1

- (a) The start of transcription of the gene *IFNA2* by *E. coli* with the recombinant plasmid shown in Fig. 2.1 needs to be controlled to obtain an optimum yield of IFN- α .

Scientists investigated the effect of two inducers of transcription on the production of recombinant IFN- α :

- lactose, which is converted to allolactose in *E. coli*
- IPTG, which is a synthetic molecule with a very similar structure to allolactose. IPTG **cannot** be broken down by *E. coli*.

The scientists grew three cultures of *E. coli* containing the recombinant plasmid in the same growth medium. The growth medium contained glucose, amino acids, essential vitamins and minerals. The growth medium did **not** contain lactose.

After four hours, either lactose or IPTG at the same concentration was added to two of the cultures of *E. coli*. As a control, the third culture of *E. coli* was grown without adding lactose or IPTG.

The concentration of recombinant IFN- α in the cultures was measured at different times over a period of 28 hours. The results are shown in Fig. 2.2.

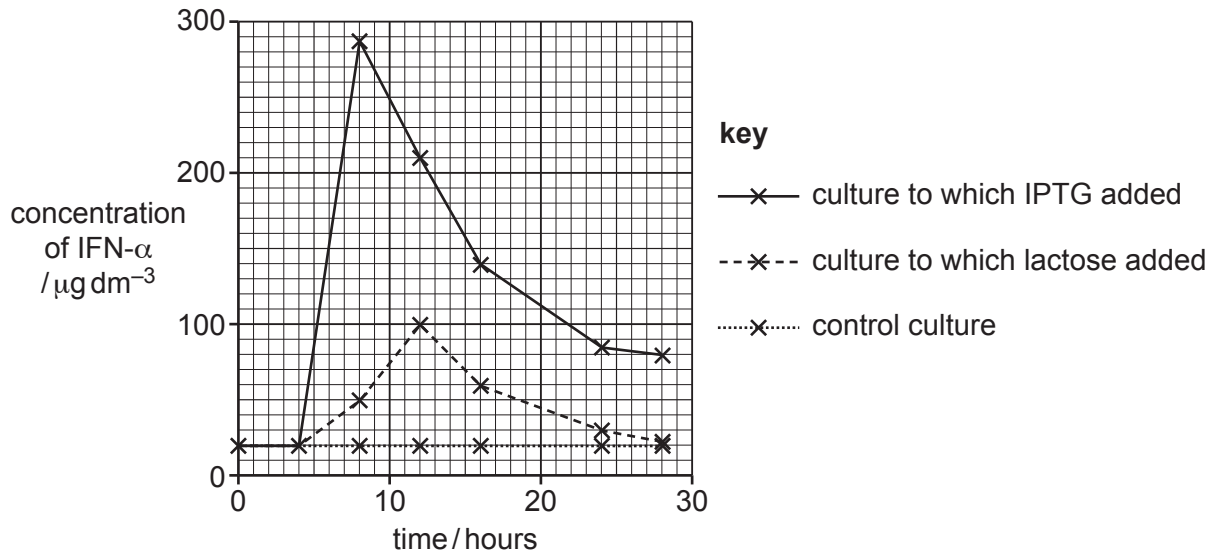


Fig. 2.2

- (i) The regulatory sequences of the *lac* operon contained in the recombinant plasmid are involved in the control of transcription of the gene *IFNA2*.

Explain the role of the gene *lacI* in the control of transcription of the *IFNA2* gene between **0 hours** and **4 hours**.

.....

.....

.....

.....

..... [2]

- (ii) With reference to Fig. 2.2, describe the changes in the concentration of recombinant IFN- α in the culture containing IPTG from when IPTG was added at **4 hours** to the end of the experiment at **28 hours**.

.....

.....

.....

.....

.....

.....

..... [3]

- (iii) Suggest **one** reason for the difference between the concentration of recombinant IFN- α in the culture at **8 hours** in the presence of lactose and the concentration of recombinant IFN- α in the culture at **8 hours** in the presence of IPTG.

.....

.....

..... [1]

- (iv) Suggest **one** reason for the change in the concentration of recombinant IFN- α in the culture containing IPTG from **12 hours** to **16 hours**.

.....

.....

..... [1]

- (b) The gene *AMP^R* in the plasmid shown in Fig. 2.1 codes for a protein that provides resistance to the antibiotic ampicillin.

Suggest how *AMP^R* allows genetically engineered *E. coli* containing the recombinant plasmid to be identified.

.....

.....

..... [1]

(c) Bacteria can evolve antibiotic resistance through natural processes.

Outline how bacteria can evolve to become resistant to antibiotics.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 13]

Question 3 starts on page 10.